

FULL NAME: \_\_\_\_\_

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## TAYLORS COLLEGE SYDNEY

### HSC MATHEMATICS

#### EXTENSION 2

30<sup>th</sup> November 2005

TIME ALLOWED: 1 hour

#### INSTRUCTIONS

1. All questions may be attempted.
2. All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
3. Approved non-programmable calculators and templates may be used.
4. Start each question on a new page.
5. Each question is to be handed in separately.

#### Question 1

Marks

(a) If  $z = 5 + 12i$  find, in cartesian form,

(i)  $\frac{\bar{z}}{zz}$  1

(ii)  $\frac{13}{z}$  2

(iii)  $\sqrt{z}$  3

(iv) the roots of  $x^2 - 3x + 1 - 3i = 0$ . 3

(b) Let  $\beta = -1 + i$

(i) Express  $\beta$  in modulus-argument form. 2

(ii) Express  $\beta^{10}$  in modulus-argument form. 2

(iii) Find the least positive integer value of  $n$  for which  $\beta^n$  is real. 1

(iv) Solve  $z^4 = 8\sqrt{2}\beta$ . 3

Express your answers in modulus- argument form.

(c) Sketch, on an Argand diagram, the locus of  $z$  where 2

$$|z + 3 - 4i| = 5$$

Question 2 Start a new Answer Booklet

Marks

- (a) Shade the region on an Argand diagram where the inequalities

$$|z| \leq |z - 4 + 2i| \text{ and } |z - \bar{z}| < 4$$

hold simultaneously.

- (b) On the same Argand diagram, sketch the locus of  $z$  where

(i)  $\arg(z - 2) = \arg(z - 3 + i)$

(ii)  $\arg(z + 1 + i) = \frac{\pi}{4}$

- (iii) Find  $z$  where  $\arg(z - 2) = \arg(z - 3 + i)$  and  $\arg(z + 1 + i) = \frac{\pi}{4}$  intersect.

- (c) (i) Solve  $z^3 = -1$ . Give your answers in modulus - argument form.

- (ii) If  $\alpha$  is a complex root of  $z^3 = -1$ , find the value of  $\alpha^2 - \alpha + 1$ .

- (iii) Hence evaluate  $(1 - \alpha)^6$ .

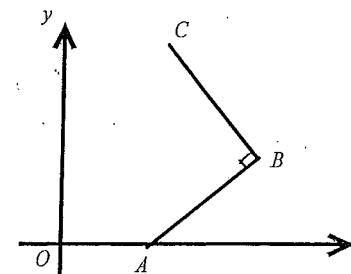
Question 3 Start a new Answer Booklet

- (a) Solve  $z^2 + 2\bar{z} + 6 = 0$ .

Question 3 Continued

Marks

- (b) The Argand diagram shows the fixed points A, B and C where  $AB = BC$  and  $\angle ABC = 90^\circ$ . The points A and B represent the complex numbers  $a$  and  $2a + bi$ , where  $a$  and  $b$  are positive constants.



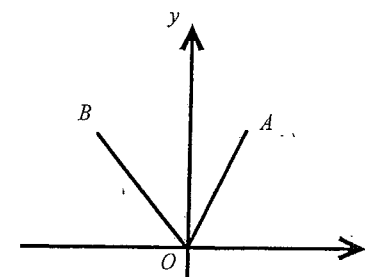
- (i) Find the complex number represented by the vector  $\overrightarrow{BA}$ .

- (ii) Find the complex number represented by the vector  $\overrightarrow{BC}$ .

- (iii) Find the complex number represented by C.

- (iv) If ABCD is a square, find the complex number represented by D.

(c)



The points A and B represent the complex numbers  $z_1$  and  $z_2$  respectively.  $\triangle ABO$  is equilateral where O is the origin.

- (i) Write  $z_2$  in terms of  $z_1$ .

- (ii) Show  $z_1^2 + z_2^2 = z_1 z_2$ .

- (iii) Deduce that if  $z_1, z_2$  and  $z_3$  are the vertices of any equilateral triangle in the complex number plane then  $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$

# Maths Extension 2

T4 2005

Q1. (a) (i)  $z\bar{z} = (5+12i)(5-12i)$   
 $= 25 + 144$   
 $= 169$

(1) ans

OR  $z\bar{z} = |z|^2$   
 $= (\sqrt{5^2 + 12^2})^2$   
 $= 169$

(ii)  $\frac{13}{z} = \frac{13}{5+12i} \times \frac{5-12i}{5-12i}$   
 $= \frac{13(5-12i)}{169}$   
 $= \frac{5}{13} - \frac{12}{13}i$

(1) "rationalizing  
denom"

(1) ans

(iii) let  $(a+ib)^2 = 5+12i$  where  $a, b$  are real

Equating real & Imaginary Parts

$a^2 - b^2 = 5$  -- (1)

$2ab = 12$  -- (2)

(1) simult. eqs

from (2)  $b = \frac{6}{a}$

sub in (1)  $a^2 - \left(\frac{6}{a}\right)^2 = 5$

$a^4 - 5a^2 - 36 = 0$

$(a^2 - 9)(a^2 + 4) = 0$

$a^2 = 9$

$(a^2 \neq -4 \text{ as } a$

$a = \pm 3$

is real)

(1) value of  $a$

$\therefore \sqrt{z} = \pm (3+2i)$

(1) ans.

(iv)  $z = \frac{2i \pm \sqrt{(2i)^2 + 4 \cdot 2(3+6i)}}{4}$

sub in  
(1) quad  
formula

$= \frac{2i \pm \sqrt{-4 + 24 + 48i}}{4}$

$= \frac{2i \pm \sqrt{20 + 48i}}{4}$

$= \frac{2i \pm 2\sqrt{5+12i}}{4}$

$= \frac{i \pm \sqrt{5+12i}}{2}$

(1) eval.  $\sqrt{\phantom{x}}$

$= \frac{i \pm (3+2i)}{2}$

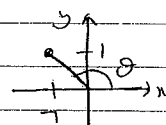
from (iii)

$= \frac{3+3i}{2} \text{ or } \frac{-3-i}{2}$

(1) ans.

b) (i)  $|-1+i| = \sqrt{1^2 + 1^2} = \sqrt{2}$

(1) mod



$\tan \theta = -1$

$\theta = \frac{3\pi}{4}$

(1) argument

$\therefore -1+i = \sqrt{2} \text{ cis } \frac{3\pi}{4}$

(ii)  $\beta^{10} = \left(2^{\frac{1}{2}} \text{ cis } \frac{3\pi}{4}\right)^{10}$

$= 2^5 \text{ cis } \frac{15\pi}{2}$

by de Moivre's thm (1)

$= 32 \text{ cis } \frac{3\pi}{2}$

$= 32 \text{ cis } (-\frac{\pi}{2})$

(1) ans

$$(III) \rho^n = \left( 2^{\frac{n}{2}} \cos \frac{3\pi}{4} \right)^n$$

$$= 2^{\frac{n^2}{2}} \cos \frac{3n\pi}{4} \quad \text{by de Moivre's}$$

If  $\rho^n$  real then  $\text{Im}(\rho^n) = 0$

$$\therefore 2^{\frac{n^2}{2}} \sin \frac{3n\pi}{4} = 0$$

$$\sin \frac{3n\pi}{4} = 0$$

When  $n=4$ ,  $\sin 3\pi = 0$

$\therefore$  least pos. integer is 4.

(k) for statement

(V) answer

$$(IV) z^4 = 8\sqrt{2} \cdot \sqrt{2} \cos(3\pi/4 + 2k\pi)$$

$$= 16 \cos \frac{\pi}{4} (3 + 8k)$$

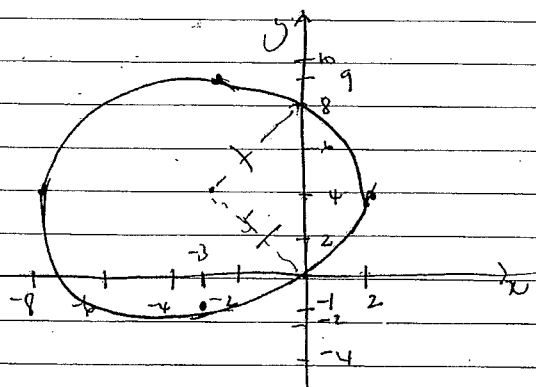
$$z = 16^{1/4} \cos \frac{\pi}{4} (8k + 3)$$

(1) de Moivre

$$z = 2 \cos \frac{\pi}{16} (8k + 3), \quad k = 0, \pm 1, -2 \quad (1) \text{ 4 solns}$$

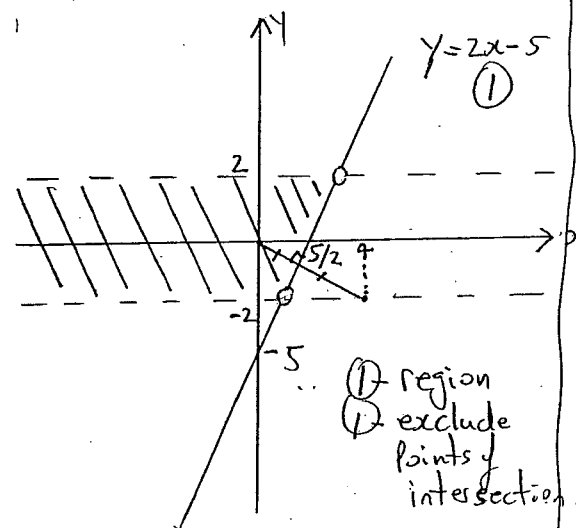
$$z = 2 \cos \frac{3\pi}{16}, \quad 2 \cos \frac{\pi}{16}, \quad 2 \cos \frac{-5\pi}{16}, \quad 2 \cos \frac{-13\pi}{16}$$

9) Circle centre  $(-3 + 4i)$  radius = 5. (1)

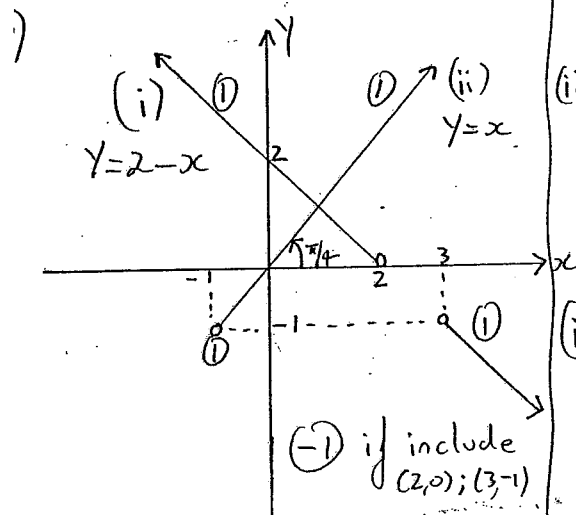


(1) passes through D or domain + range correct

Question 2 Ex2 Term 4 '05



$$2 - \bar{z} \leq 4 \rightarrow |y| \leq 2 \quad (1)$$



$$(iii) (1, 1) \rightarrow z = 1 + i \quad (1)$$

$$c) z^3 = -1$$

$$(i) |z^3| = |z|^3 = |-1| = 1$$

$$\therefore |z| = 1$$

Let  $\arg z = \theta$

$$\therefore \arg z^3 = 3\arg z = \arg(-1)$$

$$\therefore 3\theta = \pi, 3\pi, 5\pi$$

$$\text{so } \theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3} \quad (1)$$

$$\therefore z = \cos \frac{k\pi}{3}, \quad k = 1, 3, 5$$

or

$$= \cos \pm \pi/3, \cos \pi$$

$$(ii) (z+1)(z^2-z+1)=0$$

If  $z$  is a complex root then  $z^2 - z + 1 = 0 \quad (1)$

$$(iii) \text{ From (ii) } 1 - z = -z^2$$

$$\therefore (1-z)^6 = (-z^2)^6$$

$$= (-1)^6 (z^2)^6$$

$$= (z^{12})$$

$$= (z^3)^4$$

$$= (-1)^4$$

$$= 1 \quad (1)$$

3a.  $(x+iy)^2 + 2(x-iy) + 6 = 0$ ,  $x, y \in \mathbb{R}$ ,  $z = x+iy$   
 $x^2 - y^2 + 2x + 6 = 0$   
 $2xy - 2y = 0$  (1)

$y(x-1) = 0 \Rightarrow y = 0$  or  $x = 1$ .

If  $y = 0$ ,  $x = \frac{-2 \pm \sqrt{4-24}}{2} = -1 \pm i\sqrt{3}$ , but  $x \in \mathbb{R} \therefore y \neq 0$  (1)  
 $\therefore x = 1$  &  $-y^2 + 1 + 2 + 6 = -y^2 + 9 = 0 \therefore y = \pm 3 \therefore z = 1 \pm 3i$  (1)

b(i)  $\vec{OB} = \vec{OA} + \vec{AB} = a + \vec{AB} = 2a + bi \therefore \vec{BA} = -\vec{AB} = -(2a - a + bi) = -a - bi$  (1)

(ii)  $\vec{BA} = \vec{BC} i - \vec{BC} = \vec{BA} / i = -\vec{BA} i = -(-a - bi)i = -b + ai$  (1)

(iii)  $\vec{OC} = \vec{OB} + \vec{BC} = 2a + bi + -b + ai = 2a - b + (a + b)i \therefore C$  represents  $2a - b + (a + b)i$  (1)

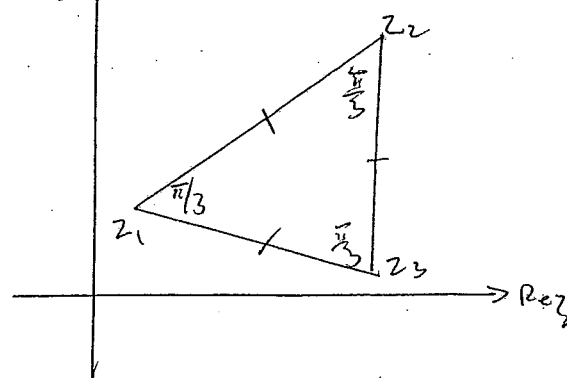
(iv)  $\vec{OD} = \vec{OA} + \vec{AB} = \vec{OA} + \vec{BC} = a + (-b + ai) = (a - b) + ai \therefore D$  represents  $(a - b) + ai$ . (1)

c(i)  $z_2 = z_1 \operatorname{cis} \frac{\pi}{3}$  (1)

(ii)  $z_1^2 + z_2^2 = z_1^2 + z_1^2 \operatorname{cis} \frac{2\pi}{3} = z_1^2 \left(1 + \frac{1}{2} + \frac{\sqrt{3}i}{2}\right) = z_1^2 \left(\frac{3}{2} + \frac{\sqrt{3}i}{2}\right) = z_1^2 \operatorname{cis} \frac{\pi}{3} = z_1 z_2$  (1)

(iii) w.l.o.g.,  $z_3 - z_1$  &  $z_2 - z_1$  satisfy the conditions in (i) & (ii)  $\therefore z_3 - z_1 = (z_2 - z_1) \operatorname{cis} \frac{\pi}{3}$  and  
 $(z_3 - z_1)^2 + (z_2 - z_1)^2 = (z_3 - z_1)(z_2 - z_1)$  (1)  
 $\therefore z_3^2 - 2z_3z_1 + z_1^2 + z_2^2 - 2z_2z_1 + z_1^2 = z_3z_2 - z_1z_2 - z_3z_1 + z_1^2$  (1)  
 $\therefore z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$ .

Q3 c) (iii)



$z_2 - z_1 = (z_3 - z_1) \operatorname{cis} \frac{\pi}{3}$

$z_1 - z_3 = (z_2 - z_3) \operatorname{cis} \frac{\pi}{3}$

$\therefore \frac{z_2 - z_1}{z_1 - z_3} = \frac{z_3 - z_1}{z_2 - z_3}$

$z_2^2 - z_2z_3 - z_1z_2 + z_1^2z_3 = z_1^2z_3 - z_1^2z_2 - z_3^2z_1 + z_3^2z_2$

$\therefore z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

or

$z_2 - z_3 = (z_1 - z_3) \operatorname{cis} \left(-\frac{\pi}{3}\right)$

$z_2 - z_1 = (z_2 - z_3) \operatorname{cis} \left(-\frac{\pi}{3}\right)$

$\frac{z_2 - z_3}{z_2 - z_1} = \frac{z_1 - z_3}{z_2 - z_3}$

$z_2^2 - z_2z_3 - z_2z_1 + z_1^2z_3 = z_1^2z_2 - z_1^2z_3 - z_3^2z_1 + z_3^2z_2$  or

$\therefore z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

Note: Alternative solution to (c)(ii) provided by LIU Long Hua (Henry):

$z_2^3 = z_1^3 \operatorname{cis} \pi = -z_1^3 \therefore z_1^3 + z_2^3 = 0$

$\therefore (z_1 + z_2)(z_1^2 - z_1z_2 + z_2^2) = 0$

$z_1 \neq -z_2$

$\therefore z_1^2 + z_2^2 = z_1z_2$ .